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# Determinants of student burnout among undergraduate private university students in Bangladesh: a structural equation modeling

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## ABSTRACT

Student burnout is a growing concern among undergraduate students. This study investigates the influence of Self-Evaluation Traits (SET), Career Decision-Making Difficulties (CDMD), and Social Support (SS) on student burnout, guided by the Conservation of Resources (COR) theory. A cross-sectional research design was employed, with data collected through convenience sampling from 401 undergraduate students across ten private universities in Dhaka city, Bangladesh. Structural equation modeling was used to examine the hypothesized relationships. The findings reveal that both SET and CDMD significantly contribute to student burnout. Social Support also has a significant direct effect on burnout. However, the interaction between SET and Social Support does not significantly influence student burnout. Social Support partially mediates the relationship between CDMD and student burnout, highlighting its importance in alleviating burnout among students facing career decision challenges. No mediating effect of Social Support was found in the SET–burnout relationship. This study contributes to the limited literature on undergraduate student burnout in Bangladesh and underscores the importance of strengthening social support systems. The findings offer valuable insights for academic policymakers and educational institutions seeking to reduce burnout and improve student well-being.

## ARTICLE HISTORY

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## KEYWORDS

Student burnout; self-evaluation trait; career decision making difficulties; social support; undergraduate students

## SUBJECTS

Social Sciences; Behavioral Sciences; Cognitive Psychology; Reading, Psychology of; Dyslexia; Attention; Mental Health; Psychological Disorders; Learning Difficulties and Mental Retardation in Children; Language, Psychology of; Developmental Psychology; Perceptual Development; Bilingualism – Second Language; Cognitive Development; Language Development; School Psychology; Assessment and Testing

## 1. Introduction

In the recent times, psychological health has become a great concern around the globe. The immense pressures that students face around the world increase the likelihood of developing severe health and psychological issues. This issue has become increasingly widespread among students, exhibiting a high incidence and severity (Wang et al., 2021). Specifically, undergraduate students at universities are often engaged in several academic responsibilities, tasks and academic deadlines. Moreover, they experience significant stresses because of external demands, high pressures, and expectations imposed by family, peers and even themselves (Hwang & Kim, 2022). The World Health Organization (WHO, 2014) reported that over 6% (450 million) of the global population experiences several mental health conditions, which comprise up to 15% of all acknowledged disorders worldwide (Sakellari et al., 2011). MacKean (2011) and Gallagher (2008) mentioned that university students, on average, experience subsequent mental

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sufferings such as anxiety, addictions, suicidal intention, and other chronic psychiatric diseases more than the average population. This may lead to the development of mental health illnesses, specifically referred to as academic burnout (Vizoso et al., 2019).

The general perception was that burnout could just affect only working individuals and professionals. It has recently been established that students may also suffer from burnout (Balogun et al., 1996; Gold & Michael, 1985; Pienaar & Sieberhagen, 2005), highlighting the necessity of investigating this problem within the context of college and university students.

In Bangladesh, as in many developing nations, substantial adjustments in higher education have occurred during the past few generations. Private universities emerged in the 1990s due to public institutions' inability to satisfy market demand and their inadequate governance (Rabbani & Chowdhury, 2014). Currently, Bangladesh comprises 53 public universities, 107 private universities, and 3 international universities with about one million students enrolled in undergraduate, graduate, and doctoral programs (University Grants Commission of Bangladesh, 2024). Prior research on the mental health of private university students in Bangladesh indicated that 60% of the pupils were suffering from anxiety, sadness, and panic attack, with poor academic scoring and workload identified as important indicators (Hoque, 2015).

Several studies have attempted to investigate the psychological well-being and contributing factors exacerbating mental health issues among students in both public and private universities, as well as burnout related to university entrance examinations or among students in other countries. In Southern Asia, university students are at a high risk of having mental health disorders. According to Dessauvage et al. (2022) students encounter depression in 29.4%, anxiety in 42.4%, and stress in 16.4%. Bangladesh has faced similar situation throughout the years. The frequency of mental health difficulty among the general people estimated at 16.1% for adulthood and 5.2% for kids and teenagers who aged 5–15 years (Islam, 2024). Previously, many health-related accidents reported among Bangladeshi university students, including self-harm, attempted suicides and suicides (Tasfi & Mostofa, 2024). Studies conducted in Bangladesh, Egypt, and Sri Lanka stated that as compare to male, female students are more likely to experience more distress (Koly et al., 2021). The Stress gradually increases with the courses of the academic year (Koly et al., 2021; Wickramasinghe et al., 2019).

While numerous international researches explore how psychological variables are link to student burnout, the findings are often conflicting, the need for specific analysis, especially in Bangladesh's private university setting. Previous research has linked self-evaluation traits (such as self-esteem, self-efficacy, and self-concept) to academic burnout, though results remain mixed. For instance, during the COVID-19 pandemic in Indonesia, nursing students exhibiting moderate self-efficacy nonetheless experienced alarmingly high exhaustion and burnout, suggesting that self-efficacy alone may not always confer protection under extreme stress (Rohmani & Andriani, 2021). Another research on core self-evaluations (CSE) indicates that students with high CSE tend to report lower burnout and are likely to leverage their personal resources, such as psychological capital and proactive coping to combat stress (Paloş, 2024).

Similarly, career decision-making Difficulties (CDMD) have shown indirect or conditional effects on burnout. For example, students who identify strongly with their profession tend to have higher academic self-efficacy, which in turn lowers burnout levels a partial mediating effect observed in music education majors (Zhu, 2025). Meanwhile, parental interference in students' career decisions has been associated with higher levels of school burnout and depressive symptoms, suggesting approach towards career choice externally imposed may impact well-being (Angelini et al., 2025).

Although social support is widely recognized as a protective factor against burnout but it showed some conflicting results. However, some studies find that social support doesn't directly reduce burnout but does so indirectly, by enhancing resilience (Liu & Cao, 2022). Moreover, Ye et al. (2021) claimed that social support that was negatively associated with academic burnout. This mixed finding stated a critical gap that no study has specifically examined how self-evaluation traits, career decision-making difficulties, and specially taking social support as a mediating variable, interact to predict academic burnout among undergraduate students at private universities in Bangladesh.

The present study explores whether self-evaluation traits and career decision-making difficulties are significant predictors of academic burnout and whether social support mediates these relationships in

the Bangladeshi private university context. Therefore, this study asks: Do self-evaluation traits and career decision-making difficulties significantly predict academic burnout, and does social support mediate these relationships, among undergraduate students at private universities in Bangladesh? By focusing on this specific demographic, the current study aims to uncover the psychological and social traits underlying academic burnout in Bangladesh's higher education system and inform targeted prevention and policy interventions.

## 2. Literature review and hypotheses development

### 2.1. *The Conservation of Resources (COR) theory*

The Conservation of Resources (COR) theory (Hobfoll & Freedy, 1993) provides a fundamental framework for comprehending the root causes of burnout and the reactions associated with sustained work-related stress. COR theory highlights the circumstances like physical fatigue or overwhelming workload can erode individuals' confidence in their ability to engage and maintain motivation, underscoring the theory's focus on the effects of resource depletion during stressful periods. Numerous prior researches have applied the COR theory to shed light on the occurrence of burnout (Alarcon et al., 2011; Huang et al., 2024; Jacobs & Dodd, 2003; Lee & Ashforth, 1996).

### 2.2. *Student burnout*

Burnout which was introduced by Freudenberg in the 1970s and gaining attention from scholars, is characterized as 'emotional exhaustion'. Schaufeli, Martínez, et al. (2002) characterized academic burnout as a state of tiredness resulting from increased academic expectations and pressures. Later, the World Health Organization redefined burnout as a syndrome arising from unmanaged chronic occupational stress (WHO, 2019).

The prevailing model of burnout, established by Maslach and Jackson (1982), characterizes burnout as a psychological syndrome resulting from persistent interpersonal stressors in the workplace, identified by three dimensions that are emotional exhaustion, depersonalization, and a reduced perception of completion in higher education. Emotional exhaustion denotes the state of being exhausted and drained of emotional resources (Lazarus & Folkman, 1984). Depersonalization is defined by dysfunctional emotional detachment, often arising from emotional tiredness, as a means to create distance between oneself and service consumers by deliberately disregarding the attributes that makes them distinctive and engaging individuals. This refers to the interpersonal dimension of burnout. The third component relates to diminished personal accomplishment, characterized by a perception of one's contributions as ineffectual and unproductive, leading to a reduced sense of self-efficacy (Bandura, 1977).

The Maslach Burnout Inventory-Student Survey (MBI-SS; Schaufeli, Martínez, et al., 2002; Schaufeli, Salanova, et al., 2002) is widely used to measure burnout in higher education. The MBI-SS was developed to evaluate the three primary symptoms of burnout within higher education (Schaufeli, Salanova, et al., 2002), which detects three subscales of burnout based on the following symptoms: emotional exhaustion, cynicism, and diminished accomplishment.

Recently, burnout among students has been connected with heavy academic workload, performance pressure, and inadequate coping strategies (Chong et al., 2025). Course load, midterm and final examinations, demonstrations or vivas, temporary employment, apprenticeships, family pressure, and hands-on experience for students with deadlines are more common at Bangladeshi private colleges than in public universities (Galbraith & Merrill, 2015). A large segment of education has switched to online and distance education, particularly during the COVID-19 pandemic. Longer time on screens, tablets, and smart devices has also resulted in increased levels of stress and burnout (Mheidly et al., 2020). Therefore, excessive burnout can result in elevated absenteeism, diminished motivation to finish coursework, a greater drop-out rate, and ultimately impact overall academic performance (Yang, 2004). This indicates that individuals become incapable of doing their duties due to excessive workload (Akbolat et al., 2010; Cooper et al., 2017). Burnout induces feelings of strain, overwhelm, and exhaustion in pupils. This consequently

adversely affects student performance, enjoyment, and diminishes productivity in learning (Norez, 2017; Sugara et al., 2020).

In the current research, student burnout is examined as the primary dependent variable, influenced by personality traits (self-evaluation), decision-making processes (career decision-making difficulties), and the mediating role of social support. Understanding its predictors can inform interventions to improve student well-being and learning outcomes in higher education.

### 2.3. Self-evaluation trait and student burnout

Core self-evaluations constitute a fundamental examination of an individual's worth, efficacy, and competence. Judge et al. (1997) proposed the core self-evaluations trait as a valuable indicator that related to work satisfaction and other associated factors. Judge et al. (1997) define core self-evaluations as a comprehensive, higher-level trait characterized by four characteristics in personality literature like (a) self-esteem, a general evaluation of oneself (Harter, 1990); (b) self-efficacy, an assessment of one's ability to perform across diverse situations (Locke et al., 1996); (c) neuroticism, the propensity for a negative cognitive/explanatory style and a focus on adverse self-aspects (Watson & Tharp, 1989) and (d) locus of control, beliefs regarding the causes of life events, with an internal locus indicating that individuals perceive events as contingent upon their own actions (Rotter, 1966).

From a theoretical viewpoint, students with high CSE are more intended to perceive academic challenges as manageable, thereby conserving psychological resources and reducing susceptibility to burnout, consistent with the Conservation of Resources (COR) theory (Holmgreen et al., 2017).

Moreover, empirical studies confirm this relationship. Dahlin et al. (2007) found that performance-oriented self-esteem was associated with burnout among medical students. According to the study by Bresó et al. (2011) higher self-efficacy increased student engagement, which in turn improved performance and lowered burnout risk. Recent research strengthens the connection between them. Yu (2024) claims that among high school students, mindfulness interventions reduced academic burnout, with CSE serving as a mediating variable, accounting for approximately 15.8% of the total effect. Similarly, pediatric residents in western China showed a negative correlation between self-efficacy and burnout where resilience mediated this relationship where self-efficacy helps protect against exhaustion (Du et al., 2024). These findings suggest that individuals with higher CSE that means stronger self-esteem and self-efficacy, experience lower levels of burnout. Therefore, we can propose the first hypothesis:

*H1a: Self-Evaluation Trait has a significant impact on student burnout.*

### 2.4. Career decision-making difficulties and student burnout

The concept of professional indecision possesses to the difficulties individuals encounter in choosing career decision (Slaney, 1988). Gati et al. (1996) introduced a taxonomy for career decision-making challenges to discern the specific obstacles encountered by individuals in the decision-making process. The challenges are categorized into sub-groups: absence of readiness, insufficient information, and incoherent data (Gati et al., 1996). The group of lack of readiness is subdivided into three distinct groups of challenges that may arise prior to the career decision-making process: (1) absence of motivation to engage in the career decision-making process; (2) indecision relating to different kinds of choices; and (3) dysfunctional ideologies, along with unreasonable demands related to the career decision-making process. Challenges may emerge in the other two disciplines during the career decision-making process.

Theoretically, prolonged uncertainty and confusion in career decision-making can function as chronic stressors heading towards burnout. According to COR theory, the ongoing demand of managing these difficulties without adequate coping resources heightens exhaustion and disengagement. Several previous research results provide empirical support for this linkage. For instance, Zhang et al. (2022) linked career stress to increased career indecision in Chinese university students. Career stress negatively affected planning and increased indecision, with 'future work self' mediating the effects of both stress and career exploration. Furthermore, Zhou et al. (2024) found the connection between social support that mediate the relationship in the study to CBMD that social support indirectly reduced CDMD

through the sequential mediation of psychological capital and career decision-making self-efficacy, underscoring the role of internal and external resources in mitigating the risk of decision-making strain.

Analyzing this evidence, career decision-making difficulties can consider as a direct indicator of burnout. Students facing high levels of CDMD are more likely to report exhaustion and disengagement, especially when lacking supportive resources such as social support, resilience, or psychological capital. Hence, the second hypothesis is proposed:

*H1b: Career decision-making difficulties have a significant impact on student burnout.*

## 2.5. Mediating role of social support

Social support is characterized by the presence of individuals that depended upon and who demonstrate concern and appreciation for others (Sarason et al., 1983). According to conservation of resources (COR) concepts, social support functions as an essential resource that buffers stress by replenishing depleted emotional and psychological reserves (Hobfoll, 1989).

Research on burnout and social support indicates that social support has consistently shown a negative relationship with burnout. According to Beltrán et al. (2009), individuals with higher social support reported lower emotional exhaustion and greater personal accomplishment. In contrast, Kutsal (2009) observed that students with low social support exhibited higher school burnout, while Jacobs and Dodd (2003) reported a negative correlation between peer support and burnout among private school students. However, Zhang et al. (2024) examined that stress can reduce social interactions, thereby diminishing available support.

Several studies illustrate social support as a mediate role. According to Gao et al. (2025), social support mediated the relationship between participation of sports and academic burnout, addressing social support that transform positive behaviors into reduced burnout among Chinese teenagers. Another study by Liu et al. (2023) confirmed that social support was found to mediate the link between stress and job burnout. In line with these findings, this study examines social support as a mediator in the relationship between self-evaluation traits, career decision-making difficulties, and student burnout. Accordingly, we can formulate the hypothesis:

*H2a: Self-Evaluation Trait has a significant impact on Social Support.*

*H2b: Career decision-making difficulties have a significant impact on Social Support.*

*H2c: Social support mediates the relationship between self- evaluation trait and student burnout.*

*H2d: Social support mediates the relationship between career decision-making difficulties and student burnout.*

*H3: Social Support has a significant impact on student burnout.*

## 3. Research materials and methods

### 3.1. Sampling design and data collection

Creswell et al. (2003) assert that a quantitative research strategy is optimal when the primary objective is to ascertain the correlations among constructs. This research employs a logical methodology, focusing on testing hypotheses developed from established theory (Bryman, 2007).

We employed the survey method to evaluate our hypothesis. A cross-sectional method is employed to gather data via offline and online self-administered survey questionnaires. The study's target population consisted of undergraduate students from private universities in three primary disciplines: business, social sciences, and science and engineering, located in Dhaka City. The current study utilized simple random sampling to collect data from the participants. Prior to participation, written informed consent was obtained from all respondents through the Google Form itself. The consent was included at the beginning of the form, where participants were provided with detailed information about the study's



**Table 1.** Demographic attributes.

| Particulars    | Descriptions            | Frequency ( <i>n</i> = 401) | Percentage |
|----------------|-------------------------|-----------------------------|------------|
| Gender         | Male                    | 233                         | 58%        |
|                | Female                  | 168                         | 42%        |
| Age            | 18–22 years             | 226                         | 56%        |
|                | 23–26 years             | 158                         | 39%        |
|                | 27–Above years          | 17                          | 5%         |
| Field of study | Business                | 196                         | 49%        |
|                | Social Science          | 53                          | 13%        |
|                | Science and Engineering | 152                         | 38%        |
| Year of study  | 1st Year                | 83                          | 20%        |
|                | 2nd Year                | 119                         | 30%        |
|                | 3rd Year                | 128                         | 32%        |
|                | 4th Year                | 71                          | 18%        |

purpose, procedures, and potential risks. Participants were informed of their right to withdraw at any stage without consequence. By selecting ‘I agree’ to proceed with the survey, participants gave their written consent to participate in the study, indicating their voluntary participation and understanding of the study’s objectives. A variety of methods was employed for data collecting, including online questionnaires and self-administered techniques. The majority of data were obtained using an online survey, with a limited number of replies gathered by printed questionnaires. Students favored the online method for completing the survey. Twenty-five replies were gathered via a printed questionnaire, revealing some missing data that were not included into the PLS software. Adhering to the guideline of 10 responses per item, we terminated the response link upon reaching 401 responses. Data were collected from July 2024 to October 2024. Reliability was assessed through internal consistency measures, including Cronbach’s alpha and composite reliability values, all of which exceeded the recommended thresholds. Validity was examined using both convergent and discriminant validity tests.

**Table 1** presents the demographic information of the respondents, which comprises 401 participants. The gender breakdown indicates a majority of males (233, 58%) compared to females (168, 42%). The vast majority of respondents, 226 individuals (56%), are aged 18–22 years, followed by 158 participants (39%) in the 23–26 years category, and a small fraction (17, 5%) aged 27 years and older. The respondents’ fields of study were predominantly Business (196, 49%), followed by Science and Engineering (152, 38%), and Social Science (53, 13%). The academic year of respondents demonstrates a varied composition: 1st-year students comprised 83 (20%), 2nd-year students 119 (30%), 3rd-year students 128 (32%), and 4th-year students 71 (18%). This breakdown of demographics assures that the research will have a diversified and representative sample, which will reflect a wide range of academic backgrounds, genders, and age groups.

### 3.2. Measures

The authors utilized a Five-point Likert scale, from 1 to 5, including ‘strongly disagree’ to ‘strongly agree’, to assess all applied constructs. This study has employed constructs from previous literature with minor modifications to ensure contextual consistency. The final questionnaire was reviewed by two specialists to ensure uniformity. A pilot test was undertaken to verify the reliability and validity of the adopted items within the Bangladeshi setting, yielding 30 responses from students. The results indicated a strong internal consistency of the constructs, with values ranging from 0.79 to 0.93, so adequately meeting the 0.7 threshold established by Hair et al. (2019). The self-evaluation trait was assessed using 12 items from Judge et al. (1997). Statements such as ‘I am assured I will attain the success I merit in life’. The assessment of Career Decision Making Difficulty was conducted utilizing the eight items from Gati et al. (1996). We utilized statements such as ‘I possess a comprehensive understanding of my career interests and strengths’. Furthermore, social support components were sourced from Zimet et al. (1988). Inquiries such as ‘I am able to discuss my issues with my family’. Ultimately, Student Burnout metrics were derived from Schaufeli, Martínez, et al. (2002). Statements like ‘I feel emotionally exhausted by my academic pursuits’.

### 3.3. Data analysis

We conducted data analysis utilizing IBM SPSS version 26 and Smart PLS-4, employing partial least squares structural equation modeling (PLS-SEM). This research employed PLS-SEM due to the following considerations. This tool is highly used when the research aims to anticipate and investigate endogenous variable to elucidate the maximum variance. PLS-SEM is the most efficient and effective prediction-focused methodology (Roldán & Sánchez-Franco, 2012). It can concurrently manage both the measurement (outer) and structural (inner) models of the framework. Moreover, it constitutes a suitable methodology for the examination of intricate path models (Hair et al., 2016). Ultimately, PLS-SEM accommodates tiny sample sizes and yields more precise results. Consequently, PLS-SEM seems to be a suitable methodology for the research.

## 4. Results

### 4.1. Common method bias test

Cross-sectional surveys occasionally ignore Common Method bias (CMB), a possible source of measurement error, especially when they use self-report questionnaires like the one used in this study. The CMB measures the difference between the observed association as established by common method variance (CMV) and the true correlation among constructs (Widodo et al., 2022). The validity and trustworthiness of the research outcomes are seriously threatened by the discrepancy between perceived and actual correlations, which is made possible by CMV (Spector et al., 2019). Fuller et al. (2016) addressed this issue by suggesting a mix of procedural and statistical enhancements to reduce and regulate CMV. The current study used few techniques to investigate this problem. The following are some examples of procedural improvements: (1) creating clear statements and different items for predictor and criteria variables; (2) adding an introductory section to the questionnaire that provides clear instructions or explanations to reassure respondents that the data acquired will be secured, protected, and utilized exclusively for research purpose; and (3) performing a pilot test with 30 participants as a first step. As recommended by Harman (1976), a single-factor statistic test was also conducted to ascertain whether CMV was present among the covariates. According to the results, all of the model's elements were divided into four latent constructs, with the first one responsible for 31.273% of the variation overall which is below cut off value of 40% assuring no presence of CMV. Finally, to identify collinearity problems, statistical methods like the full collinearity test in Smart PLS 4, as recommended by Hair et al. (2019), were employed. We conducted a comprehensive collinearity assessment test, encompassing both inner and outside models, in Smart PLS. The Variance Inflation Factor (VIF) values for each construct were analyzed, employing a threshold of 3.3 to identify the existence of collinearity. This approach is contemporary and more dependable, as indicated by Kock (2021) and endorsed by other social science scholars (Zafar et al., 2020). All VIF figures are below the cut off value of 3.3, indicating that common method bias is not an issue in our model (Kock, 2021; Tehseen et al., 2017).

### 4.2. Analysis of measurement model

The outer (reflective) measurement model was used to assess reliability, internal consistency, convergent validity, and discriminant validity for all constructs (Sarstedt et al., 2017). Convergent validity is 'the degree to which a measure correlates favorably with alternative measures of the same construct' (Hair et al., 2016) and was evaluated using indicator outer loadings and the average variance extracted (AVE). Outer loadings  $\geq 0.708$  indicate that the indicators share substantial variance with their construct, and AVE  $\geq 0.50$  indicates that the latent variable explains more than half of the variance in its indicators. For single-item constructs, the outer loading is fixed at 1.00 and AVE is not applicable. A schematic of the measurement model and outer loadings is provided in Figure 1.

The findings indicated that the scores of Composite reliability (CR) and Cronbach's Alpha (CA) exceeded the critical threshold of 0.70, as proposed by Cohen (1988). The standardized factor loading values for all items of the constructs exceed 0.70, with a few items marginally below this threshold. Additionally, the average variance extracted (AVE) for all constructs is above 0.50, indicating a



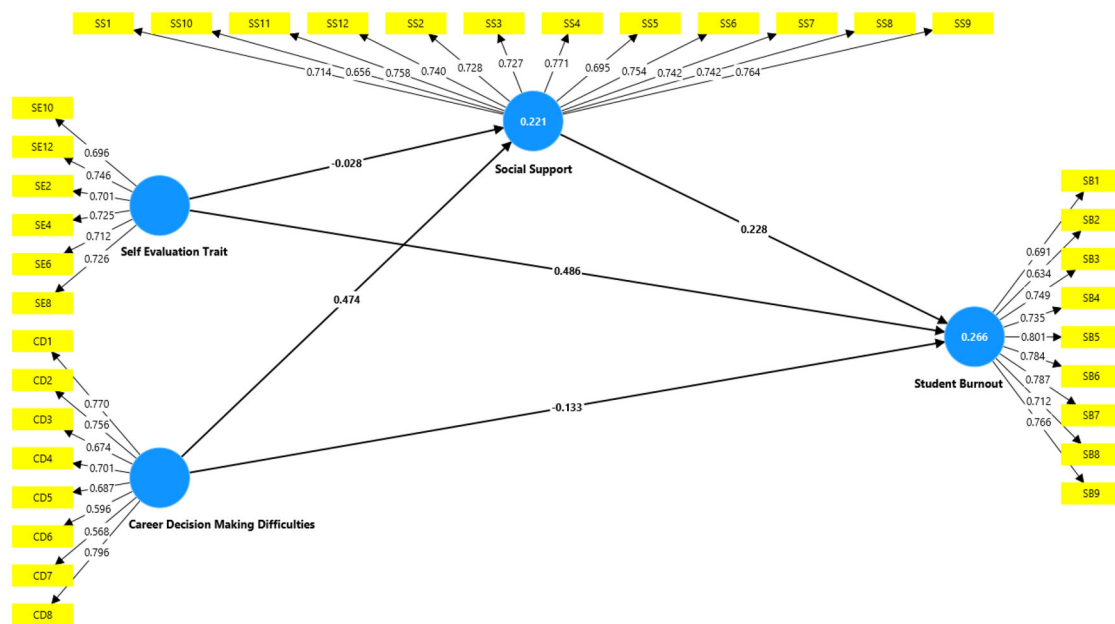


Figure 1. Measurement model with outer loadings (Source: Authors' own illustration).

Table 2. Construct validity and reliability.

| Constructs | Items | Factor loading | Cronbach's alpha | Composite reliability <sup>a</sup> | Average variance explained <sup>b</sup> (AVE) | Convergent validity (AVE > 0.5) |
|------------|-------|----------------|------------------|------------------------------------|-----------------------------------------------|---------------------------------|
| SET        | 06    | 0.696–0.726    | 0.812            | 0.864                              | 0.515                                         | Yes                             |
| CDMD       | 08    | 0.770–0.796    | 0.847            | 0.882                              | 0.507                                         | Yes                             |
| SS         | 12    | 0.714–0.771    | 0.922            | 0.933                              | 0.538                                         | Yes                             |
| SB         | 09    | 0.712–0.801    | 0.896            | 0.916                              | 0.550                                         | Yes                             |

SET = Self-Evaluation Trait; CDMD = Career Decision Making Difficulties; SS = Social Support; SB = Student Burnout. a. Composite reliability value should be > 0.7. b. Average variance extracted (AVE) value should be > 0.5.

satisfactory level of adequacy. Table 2 presents the comprehensive results pertaining to validity and reliability. The VIF was also assessed to evaluate multicollinearity across all items of measurement, yielding values below 3.3. According to Hair et al. (2019), a VIF cut-off value below 5.0 is necessary to prevent multicollinearity, while a higher value indicates a potential issue with multicollinearity. While assessing discriminant validity the study used rigorous tests. As Hair et al. (2019) explains the discriminant validity as the degree to which a construct is genuinely differentiated from other constructs based on empirical criteria. This shows that one latent construct in the model is distinct and unrepresented by other latent construct. The discriminant validity is evaluated using the indicators' cross loading. When indicators are attached to a construct, their outer loading should be bigger than the cross loadings on other indicators items. In a similar vein, discriminant validity, as defined by Sarstedt et al. (2017), shows that each latent construct is unique from other latent constructs used in the model. This study employed two primary methods to ascertain discriminant validity: the Fornell and Larcker criterion and the Heterotrait-Monotrait (HTMT) ratio. The Fornell-Larcker criterion is the first method used to evaluate discriminant validity. Hair et al. (2019) write that 'the square root of the AVE of each construct should be higher than its strongest correlation with any other construct' (p. 107) in order to meet this requirement. The square root value of AVE, as outlined by Fornell and Larcker, must exceed the correlational values of the respective constructs (Fornell & Larcker, 1981). As per the recommendation of Hair et al. (2019), the values support the discriminant validity of the reflective constructs in the model and are compatible with the Fornell-Larcker criterion. Career Decision-Making Difficulties (CDMD) had a square root of AVE of 0.698, as indicated in the results section. This was higher than the relationships with Student Burnout (SB = 0.071), Social Support (SS = 0.469), and Self-Evaluation Trait (SET = 0.200). In the corresponding rows and columns, the diagonal values consistently outperformed the off-diagonal correlations for all other constructs, including SET (0.718), SS (0.733), and SB (0.742). The Fornell-Larcker discriminant

**Table 3.** Fornell and Larcker criterion for assessment of discriminant validity.

|      | CDMD  | SET   | SS    | SB    |
|------|-------|-------|-------|-------|
| CDMD | 0.698 |       |       |       |
| SET  | 0.200 | 0.718 |       |       |
| SS   | 0.469 | 0.067 | 0.733 |       |
| SB   | 0.071 | 0.475 | 0.198 | 0.742 |

Note: SET = Self-Evaluation Trait; CDMD = Career Decision Making Difficulties; SS = Social Support; SB = Student Burnout.

**Table 4.** Heterotrait-Monotrait (HTMT) criterion for discriminant validity.

|      | CDMD  | SET   | SS    | SB |
|------|-------|-------|-------|----|
| CDMD |       |       |       |    |
| SET  | 0.273 |       |       |    |
| SS   | 0.512 | 0.118 |       |    |
| SB   | 0.148 | 0.549 | 0.217 |    |

SET = Self-Evaluation Trait; CDMD = Career Decision Making Difficulties; SS = Social Support; SB = Student Burnout. Note: Discriminant validity is established at  $HTMT_{0.85}/HTMT_{0.90}$ .

validity condition was met since each construct shared more variation with its own indicators than with other constructs (Fornell & Larcker, 1981). The HTMT ratio is the second method that was employed for a more thorough evaluation of discriminant validity. By evaluating the adjustments of indicators across the constructs, the Heterotrait-Monotrait ratio (HTMT.90) assesses the dissimilarities of the constructs. In that respect less than 0.90 is the suggested value (Rönkkö & Cho, 2022; Yusoff et al., 2020). The Heterotrait-Monotrait (HTMT) ratio of the correlations between latent constructs was introduced by Henseler et al. (2015). The geometric mean of the average correlations for items measuring the same construct is represented by the HTMT, which denotes the mean value of item correlations across constructs. When HTMT readings are high, discriminant validity problems is noticed. The threshold value for structural models with conceptually strongly comparable constructs should be 0.90, as per Henseler et al. (2015). The conservative criteria of 0.85 and the more permissive limit of 0.90 were both significantly above the HTMT values, that ranging from 0.118 to 0.549 (Henseler et al., 2015). Between self-evaluation trait (SET) and student burnout (SB), the highest recorded HTMT value was 0.549, which is well below the suggested cut-off. This provides more evidence that Discriminant validity has been guaranteed for the latent constructs in the measurement model.

Tables 3 and 4 provide a comprehensive overview of the results. The analysis has yielded the cross loadings of the constructs, and the results demonstrate that all constructs are distinct from one another.

#### 4.3. Analysis of structural model

Examining the structural model comes next in analyzing PLS SEM data once the measurement model assessment is satisfactory. The inner or structural model shows how exogenous and endogenous constructs are causally related (Hair et al., 2016). Consistent with recommended criteria as proposed by Hair et al. (2019), we assessed the coefficient of determination ( $R^2$ ), predictive relevance via blindfolding-based cross-validated redundancy ( $Q^2$ ), the magnitude/significance of path coefficients ( $\beta$ ), and effect sizes ( $f^2$ ). The structural model with path estimates is depicted in Figure 2. Researchers can assess the effect of eliminating a specific predictor construct on the ( $R^2$ ) value of an endogenous construct. The effect size ( $f^2$ ) constitutes this statistic. The  $R^2$  value for Social Support (SS) was 0.221, while for Student Burnout (SB) it was 0.266, demonstrating the model's accurate predictive ability (Hair et al., 2016). These values show weak-to-moderate explanatory power, according to Hair et al. (2016), showing that the model contains a strong capacity for anticipating the endogenous constructs in the context of behavioral study. Additionally, we assessed the impact size  $f^2$ , which produced results for Social Support (SS) =0.005, Self-evaluation trait (SET) =0.309, and Career Decision Making Difficulties (CDMD) =0.018. According to the effect size ( $f^2$ ) results, SET (0.309) exhibits a large effect in accordance with Cohen (1988) guidelines, indicating that (SET) significantly contributes to explaining the variance in the dependent variable, while CDMD (0.018) and SS (0.005) have negligible effects

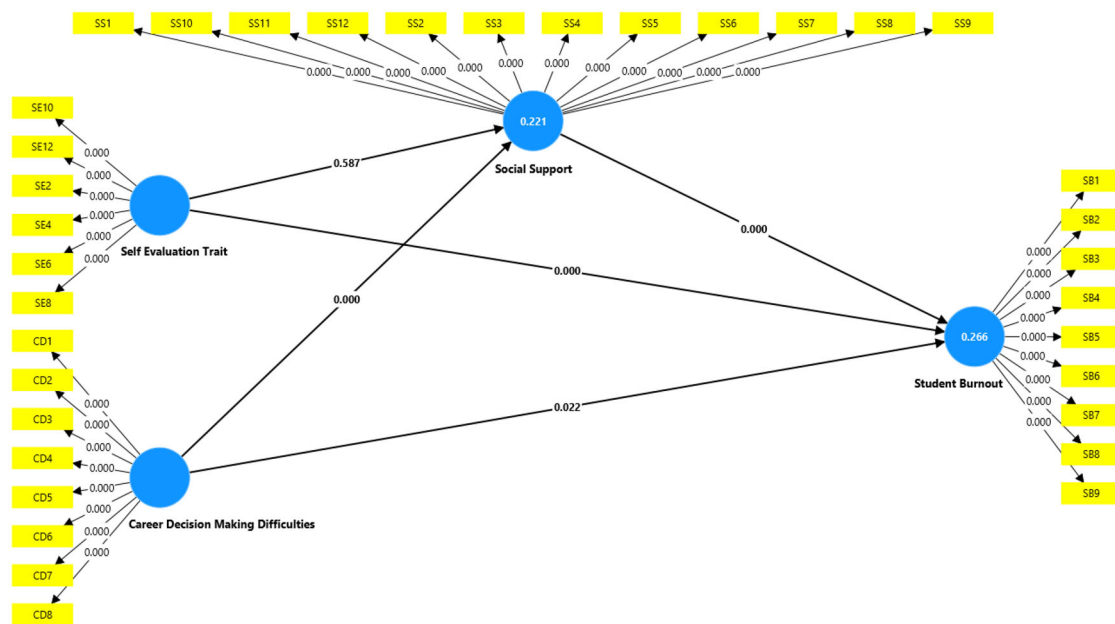


Figure 2. Structural model with  $p$ -values (Source: Authors' own illustration).

Table 5. Path co-efficient results of all direct paths to test hypotheses.

|                       | Original sample (OS-) | Sample mean (SM) | Standard deviation (SD) | $T$ -statistics | $p$ -Values | Decision      |
|-----------------------|-----------------------|------------------|-------------------------|-----------------|-------------|---------------|
| CDMD $\rightarrow$ SS | 0.474                 | 0.482            | 0.049                   | 9.688           | 0.000       | Significant   |
| CDMD $\rightarrow$ SB | -0.133                | -0.136           | 0.058                   | 2.285           | 0.022       | Significant   |
| SET $\rightarrow$ SS  | -0.028                | -0.027           | 0.052                   | 0.543           | 0.587       | Insignificant |
| SET $\rightarrow$ SB  | 0.486                 | 0.492            | 0.041                   | 11.848          | 0.000       | Significant   |
| SS $\rightarrow$ SB   | 0.228                 | 0.228            | 0.058                   | 3.941           | 0.000       | Significant   |

Notes: SET = Self-Evaluation Trait; CDMD = Career Decision Making Difficulties; SS = Social Support; SB = Student Burnout.

on their respective endogenous constructs. The blindfolding procedure yielded  $Q^2$  values of (0.44) for Student Burnout (SB) and (0.445) for Social Support (SS). Every value of  $Q^2$  exceeds zero. According to Sarstedt et al. (2017), this model exhibits a reasonable degree of predictive relevance for dependent constructs. Hair et al. (2019) state that the minimal, medium, and large predictive relevance of the PLS-path model are shown by  $Q^2$  values greater than 0, 0.25, and 0.50. The value of the standardized root means square residual (SRMR) is 0.077, which is below the threshold value of 0.080. Consequently, this serves as a confirmation of the overall adequacy of the PLS path model (Sarstedt et al., 2017). As per the formula of Goodness of fit (GoF) =  $H \sqrt{\text{Average AVE} \times \text{Average } R^2}$ , the GoF value has been determined to be 0.163, indicating a small model fit (Wetzels et al., 2009). The concept of chi-square-based model fit measures and their extensions, as utilized in CB-SEM, do not apply here since the algorithm for deriving PLS-SEM solutions does not focus on minimizing the divergence between observed and estimated covariance matrices (Hair et al., 2019). In the realm of CB-SEM studies, leading scholars have raised concerns about the applicability of the concept of model fit for PLS-SEM methodologies overall (Hair et al., 2017; Lohmöller, 1989; Rigdon, 2012). The importance of each hypothesis was evaluated using a bootstrapping method that involved 5,000 resamples. The results are detailed in Table 5, indicating that the hypothesized paths that reveal significant values, with  $T/p$  values are as follows:  $CDMD \rightarrow SS = 9.688/0.000$ ,  $CDM \rightarrow SB = 2.285/0.022$ ,  $SET \rightarrow SB = 11.848/0.000$ , and  $SS \rightarrow SB = 3.941/0.000$ . The only insignificant path is  $SET \rightarrow SS$ , which has a value of  $T/P$  0.543/0.587. Each independent variable's direct effects on the dependent variable yield noteworthy results. Post hoc analysis in regression-based systems is represented by beta value of coefficient of determination in CB-SEM based studies. The current study used Smart PLS-4 which is not CB-SEM dependent. The specific effects of each independent variable have been ascertained through original sample (beta value- $\beta$ ) and  $f^2$  values reported elaborately.

**Table 6.** Indirect effect report (mediation).

|                | OS     | SM     | SD    | UCL 2.50% | LCL 97.50% | T     | <i>p</i> Values | Decision          |
|----------------|--------|--------|-------|-----------|------------|-------|-----------------|-------------------|
| CDMD > SS > SB | 0.108  | 0.109  | 0.027 | 0.051     | 0.160      | 3.968 | 0.000           | Partial mediation |
| SET > SS > SB  | −0.006 | −0.006 | 0.013 | −0.034    | 0.016      | 0.513 | 0.608           | No mediation      |

Notes: SET = Self-Evaluation Trait; CDMD = Career Decision Making Difficulties; SS = Social Support; SB = Student Burnout; OS = Original Sample; SM = Sample Mean; SD = Standard Deviation; UCL = Upper Confidence Limit; LCL = Lower Confidence Limit.

#### 4.4. Mediation analysis

The mediating effect of Social Support (SS) was examined using the two-step procedure proposed by Hair et al. (2016). This study initially evaluated the specific indirect effects of Career Decision Making difficulties (CDMD) on Social Support (SS) and Student Burnout (SB) ( $T/p = 3.968/0.000$ ), as well as (SET) on (SS) and (SB) ( $T/p = 0.513/0.608$ ). In the subsequent step, we assessed the direct effects of (SET) and Career Decision Making difficulties (CDMD) on Student Burnout (SB), revealing that the Career Decision Making Difficulties (CDMD) to Student Burnout (SB) pathway is significant, while the mediated pathway Career Decision Making Difficulties (CDMD) to Social Support (SS) to Student Burnout (SB) also demonstrates significance, indicating partial mediation. No mediation was identified in the pathway SET > SS > SB. Table 6 provides the relevant findings.

### 5. Discussion

The present study examined the influence of self-evaluation trait (SET) and career decision-making Difficulties (CDMD) on student burnout (SB), with social support (SS) as a potential mediator, within the context of undergraduate students in private universities in Bangladesh.

Indecision in career choice constitutes a major psychological stress, draining students' emotional and motivational condition, as we observed that career decision-making challenges considerably lead to student burnout. According to earlier research, students who struggle with career decisions report higher levels of academic burnout (Olwage & Mostert, 2014); emerging adults who struggle with career decisions have been found to be more stressed and emotionally exhausted (Pham Thi & Duong, 2024); and mentoring and self-efficacy interventions have been demonstrated to lessen these decision-making challenges and lower the risk of burnout (Pham Thi & Duong, 2024). Furthermore, more comprehensive resource-based models suggest that burnout symptoms are exacerbated by a lack of internal resources and support (Juvonen & Graham, 2014). Some studies indicate that the negative effects of (CDMD) may be lessened when appropriate support networks are in place, even though the majority of research consistently demonstrates a positive correlation (Olwage & Mostert, 2014; Pham Thi & Duong, 2024). In order to prevent burnout in private university settings, our findings emphasize the need for career development treatments such organized mentoring, decision-making seminars, and self-efficacy training.

The preventive function of a student's internal confidence and positive self-assessment is highlighted by the strong correlation found between self-evaluation trait (SET) and student burnout (SB). This is consistent with the literature, which shows that strong core self-evaluations predict lower academic stress and better emotional stability are inversely linked to academic burnout and emotional exhaustion (Lian et al., 2014), and consistently predict less burnout in organizational contexts (Leupold et al., 2020). Strong self-assessment qualities appear to be internal resources that promote resistance against academic stress, according to these three convergent findings. A small number of research highlight variability based on more general contextual modifiers, despite the majority confirming this protective effect (Zhu et al., 2024). All things considered, encouraging students' self-evaluation qualities via mastery internal experiences, reflective practice, and encouraging communication and student feedback mechanism can be a potent burnout prevention tactic.

However, social support (SS) partially mediated the impact of Career Decision Making Difficulties (CDMD) on burnout, but it did not mediate the effect of (SET) on burnout. This suggests that while external support systems mitigate the negative effects of decision-making challenges, they do not eliminate them. The non-mediation pathway of Self-evaluation trait (SET) > Social support (SS) > Student burnout (SB) indicates that the protective function of SET operates more directly and independently of

external resources, rather than through perceived or activated support networks. A plausible explanation is that pupils with elevated self-evaluation features predominantly depend on their own confidence and coping mechanisms, hence diminishing the necessity for social support. The findings correlate with several research that provide data indicating that internal features can operate independently in alleviating tension (Leupold et al., 2020; Zhu et al., 2024). Academic burnout has been consistently found to be significantly correlated with social support (Ye et al., 2021); this relationship is partially mediated by life satisfaction (Ye et al., 2021); burnout symptoms are reduced by perceived support (Chong et al., 2025); and more general models of resource conservation highlight the importance of social support as an external resource that lessens exhaustion. Nonetheless, its ability to mediate may differ depending on the cultural and socioeconomic setting (Ye et al., 2021). According to this partial mediation, enhancing social support networks like peer groups and mentorship can reduce burnout associated with (CDMD), but it should be used in conjunction with other tactics like career counseling, skill-building workshops, and stress management programs rather than in isolation.

Combining these results, we find that although self-evaluation trait (SET) directly lowers burnout without any discernible mediation from social support (SS), Career Decision Making Difficulties (CDMD) directly raise burnout, but social support mitigates this effect (partial mediation). Thus, social support (SS) as external resources and self-evaluation trait (SET) as internal resources work in parallel; one attenuates harm indirectly while the other directly fights burnout. This highlights the necessity for unique intervention strategies and gives theoretical support to models that distinguish between situational and dispositional pathways to burnout. This finding emphasizes the need for universities to implement more tailored interventions to address self-evaluation deficiencies and to enhance social support (SS) systems, such as peer mentoring, faculty engagement, and family involvement, particularly during career decision-making phases

## 6. Conclusion

By tackling the urgent problem of student burnout in Bangladeshi private universities a unique setting where little research has looked at the combined impact of contextual and personal factors this study significantly advances the field. The study clarifies the human inner factors and external situational factors that contribute to burnout by examining the effects of career decision-making challenges, self-evaluation characteristics, and the mediating function of social support. The results show that self-evaluation characteristics act as a protective internal resource, professional decision-making challenges exacerbate student burnout, and social support partially mitigates the detrimental impacts of decision-making challenges but not self-evaluation. This subtle result is especially significant because it shows that internal characteristics have a more immediate and robust protective effect than external support, which can act as a buffer against some stressors. These findings are unique because they distinguish the roles of internal and external resources: self-evaluation attributes minimize burnout regardless of social support, while support systems are the only way to lessen the negative effects of career selection challenges. By implying that the mechanisms by which environmental and human resources function differently, this distinction theoretically clarifies resource-based models like the Job Demands Resources framework. While realistically highlighting the significance of creating interventions that simultaneously boost students' self-confidence and fortify institutional support networks, theoretically the research adds to the body of literature by clarifying the relative roles of internal and external elements. The study concludes that addressing student burnout necessitates a twofold approach: building organized social support networks to mitigate situational demands like job insecurity and fostering students' self-evaluative abilities to maintain resilience. Regardless of these contributions the study is limited by cross-sectional design, and concentration on undergraduates at private universities in a single country, notwithstanding these contributions. Future research should employ longitudinal designs, include diverse institutions, and explore comparative cultural contexts to validate and expand these findings. By doing so, researchers and educators can better understand how to protect students' well-being and enhance academic resilience within varying cultural and educational settings.

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## Ethics statement

Prior to data collection, ethical approval was obtained from the Institutional Ethics Committee (IEC) of Dhaka International University (DIUIEC/2024/BA-CR/02). Participation was entirely voluntary, and informed consent was secured from all respondents. Participants were briefed on the study's purpose, assured of their right to withdraw at any stage, and informed that their responses would remain strictly confidential and used solely for academic purposes. No personally identifiable information was collected.

## Authors contributions

Rahman, the primary author, contributed to the methodology, data analysis, discussion parts. Sharmen and Alam have contributed to the introduction, literature review, conclusion part and proof reading.

## Disclosure statement

The authors state that no financial or commercial relationships that could be interpreted as a potential conflict of interest existed throughout the study's execution. Authors of the study are also in consensual state of agreement.

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## Data availability statement

The information presented in this article can be obtained by reaching out to the corresponding author(s).

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